

Personal AI · Decentralized Data Science

Model Building for Numerai and Beyond with the Rise of Personal AI Agents

Alex / Aventurine · Founder - Aventurine Labs

May 30, 2026 · Prepared in Partnership with Gandalf — my OpenClaw assistant

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Alex / Aventurine

Founder · Aventurine Labs

Crypto-native consulting practice.

Numerai Council of Elders

Elected Member · Runs the public-facing CoE social account.

Background

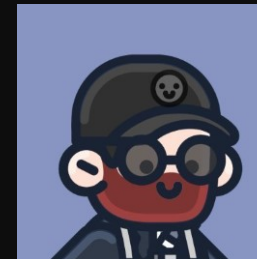
US Military → Medical Lab/Field Engineer → Finance Degree → Consulting Company

Long-running participant in decentralized AI:

Numerai, Allora, CrowdCent, CrunchDAO.

Photographer

Digital Art Collector



Find me online

X: @aventurine_eth

Personal · Crypto + AI commentary

X: @NumeraiCoE

Numerai Council of Elders Social Account

aventurinelabs.xyz

The consulting practice

If you build models or run agents or looking for consultants— say hi after the talk.



Three acts, one through-line

01

The Agent

What a personal AI agent is, how it's wired, and why local context wins.

02

The Playbook

Prompt patterns and operating habits for building models with an agent in the loop.

03

The Portfolio

Six DeAI projects I've built or hardened with my agent over the past few months.

Why this talk exists

**Personal AI agents are the new
lab assistant for the
decentralized data scientist.**

Not a chatbot. A collaborator with memory, tools, and a calendar.

Decentralized data science is broad, fast, and increasingly competitive.
You don't win it by working harder. You win it by giving yourself leverage.

The surface area exploded — and so did the leverage

- ▶ Numerai launched MCP + Skills — agents have native tournament access.
- ▶ Numerai, Allora, CrowdCent, CrunchDAO all run live competitions in parallel.
- ▶ Daily submission cadences across multiple platforms = real ops burden.
- ▶ Frontier LLMs are good enough to scaffold, critique, and self-correct.
- ▶ Personal agents persist context across weeks — humans don't.
- ▶ One operator + one agent now does what a small team did 18 months ago.

What is a personal AI agent?

Definition

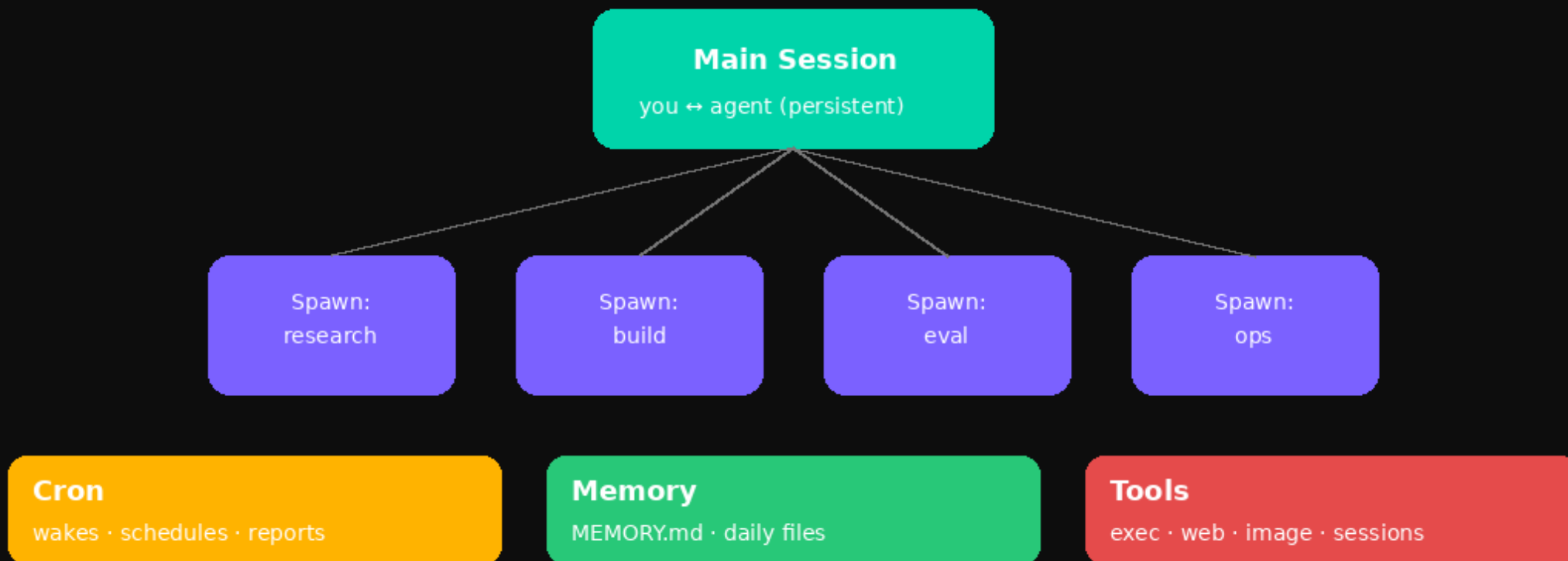
A persistent, tool-using LLM that lives on your machine, in your context, with your data — and follows your rules.

Mine is called Gandalf. Runs on OpenClaw — local-first, open source, MIT-licensed.

Not the same as a chatbot

Memory	weeks of context, not one chat
Tools	shell, web, cron, files
Identity	soul, rules, preferences
Autonomy	wakes itself; finishes tasks
Trust	your machine, your data, your rules

How the pieces fit



One main session is the door. Sub-agents are throwaway specialists. Cron is the heartbeat. Memory is the soul.

Markdown files are the agent's mind

SOUL.md

Who you are.
Tone, values, opinions,
boundaries.

Example

"Be helpful, not performative."

IDENTITY.md

What you go by.
Name, vibe, emoji, avatar.

Example

"Gandalf — the wizard."

MEMORY.md

Long-term memory.
Projects, rules, key facts.

Example

"Never stake without consent."

TOOLS.md

Local notes.
SSH hosts, API keys (gitignored),
aliases.

Plain markdown. Editable by you, readable by future agent. No vendor lock.

What the agent can actually do

exec(System Call)

Shell. The everything tool.
Run scripts, train models, deploy.

sessions

Spawn sub-agents.
Isolated, parallelized, throwaway.

cron

Schedule wake-ups.
Daily reports, retrain Sundays.

web

Search + fetch.
Docs, papers, market data.

image / vision

Generate + read charts.
Diagrams, screenshots, plots.

memory

Read/write the agent's mind.
Files are the API.

Why local + persistent compounds

Without persistence

- ▶ Generic chatbots forget you between conversations.
- ▶ Cloud agents see your queries but not your work.
- ▶ Local Training

With persistence

- ▶ Sharper recommendations week over week.
- ▶ Less repetition — say it once, write it down.
- ▶ Privacy by default — your data stays yours.
- ▶ Local cron means no vendor in the loop.

The agent gets better at helping you the longer you use it.

Prompt 1 · Deep research before you write a line of code

Deep research prompt

Before we build anything for Numerai Classic, I want you to do deep research.

1. Read every page of docs.numer.ai (Classic section).
2. Walk numerai/example-scripts on GitHub – sample notebooks, hello_numerai.py, the analysis_and_tips notebook, the AI agent example.
3. Scan the latest forum + Discord threads on the Ender20 target and v5.2 data.
4. Pull the 5 most recent NumerCon talks and summarize key model insights.

Output a 1-page brief:

- Data shape (eras, features, targets) and what's new in v5.2
- Scoring math (CORR, MMC, BMC) and current payout weights
- Top community pitfalls + the 3 model archetypes that win lately
- Sources linked inline – no claims without a URL.

Don't write code yet. Brief first.

Make the agent become the expert before you ask it to build like one.

Prompt 2 · Sourcing data and building a custom dataset

Data sourcing prompt

For Numerai Signals (bring-your-own-data), I need to build my own equity dataset.

1. Find 5 sources of free or low-cost US equity OHLCV + fundamentals.
Compare: cost, coverage, latency, license, rate limits.
2. Recommend two – one primary, one redundancy.
3. Write `fetch_universe.py`:
 - Pulls daily bars for the Numerai Signals `universe.parquet` ticker list.
 - Aligns dates/timezones, handles delistings, splits, dividends.
 - Saves partitioned parquet to `./data/raw/{source}/{date}/`.
4. Add a data-quality report: % missing per ticker, stale tickers, gaps.
5. Tell me what you would add later (alt data, intraday) – don't build it now.

Same rules for Numerai Crypto: target assets, cross-assets, OHLCV + quote volume, # trades, taker buy volume, Fear & Greed, DXY proxy, synthetic funding rate. The more features the better.

Bring-your-own-data competitions are won at the data layer, not the model layer.

Prompt 3 · Build the model + give me diagnostics

Model + diagnostics prompt

Build model.py for Numerai Classic using these rules (skip the defaults):

- Select UNIQUE feature groups – avoid the most-common feature sets.
- Train on a BLENDED target: weighted combo of 3-4 aux targets.
- Use Gradient Boosted Trees (LightGBM / XGBoost / HGBR) with bespoke tuning – not stock hyperparameters.
- Era-aware CV: purged + embargoed splits across eras.
- Neutralize predictions against top exposed features post-hoc.

Then give me diagnostics:

- Per-era CORR, MMC vs ender20 benchmark, FNC, max drawdown.
- Feature exposure plot and target-blend ablation.
- 200-era rolling CORR + Sharpe.

Save plots to ./reports/ as PNGs. Print a one-paragraph verdict at the end.

Specify the WHY (unique groups, blended targets, era-aware CV) — not the lines of code.

Where to let the agent drive

Model choice

Given the target shape and dataset size,
recommend the most boring model that works.
Ask for a fallback if it doesn't.

Feature direction

Ask for hypotheses, not features.
"What 3 themes would you test against this target?"
Then you pick what to actually build.

Evaluation metric

Don't trust a single metric.
Ask for CORR, MMC, Sharpe, drawdown,
and which one is most likely to mislead.

Schedule + cadence

When should retrain happen?
How often should it submit?
Let the agent draft, you approve.

Trust is earned, not assumed

Never stake without explicit consent.

Every. Single. Time.

Never delete models or accounts.

Ever. Under any circumstances.

Never push without asking.

Especially on shared repos.

Trash > rm.

Recoverable beats gone forever.

Ask before going external.

Emails, tweets, deploys, public posts.

Human-in-the-loop on submissions.

First run. Always.

Patterns that catch the agent's mistakes

Dry-run mode

Every pipeline has --dry-run.
Generate the file, don't submit.
Cheap to test, easy to diff.

Staged publish

Retrain to a candidate path.
Validate. Only then swap symlink to live.
Failed retrain leaves last-good untouched.

State files

state.json says "submitted today."
Reruns are idempotent.
No double-submits, no missed days.

Watchdogs

Cron checks the pipeline itself.
Process alive + work happening.
pgrep alone is never enough.

Numerai Classic

The original tournament — obfuscated equity data



What it is

- Crowd-sourced ML hedge fund
- Free, obfuscated feature data (v5.2)
- Scored on CORR + MMC vs ender20 benchmark
- Daily live data Tue–Sat

What we built — together

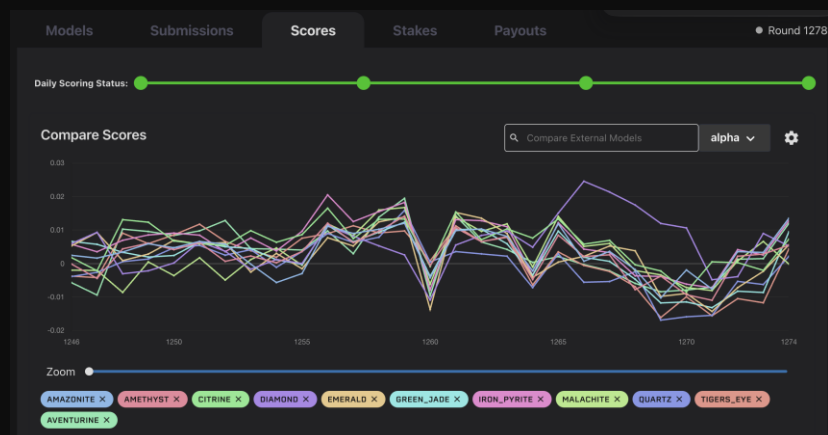
- Agent did deep research on docs + example-scripts repo
- Built training framework using Numerai's official agent kit
- Set up CV pipelines, BMC tracking, benchmark comparisons
- Validated.pkl uploads inside Docker before going live
- Now retrains and submits with human approval on first run

Tips & tricks

- ▶ Select UNIQUE feature groups — avoid the most common ones
- ▶ Era-aware CV (purged + embargoed) — never random splits
- ▶ Build BLENDED targets (weighted combo of 3-4 aux targets)
- ▶ Neutralize predictions against top-exposed features
- ▶ Use GBT with bespoke tuning — defaults are everyone's defaults
- ▶ BMC > MMC for picking experiments

Numerai Signals

Bring-your-own-data tournament — global equities



What it is

- You provide features and signals
- Scored on Alpha + MPC (neutralized, liquidity-weighted)
- Daily submissions, monthly retrains
- Payout clip raised to 3.5%

What we built — together

- Agent + I built the equity data pipeline from scratch
- Sourced OHLCV + alt data; aligned to Signals universe
- Explored multiple alpha approaches per specialty model
- Daily submission wrapper with state-aware gating
- Monthly retrain cron (first Sunday) + Telegram reporting

Tips & tricks

- ▶ Source data from at least 2 providers — redundancy matters
- ▶ Cron "fires once after fresh submission" beats time-based
- ▶ Align dates/timezones up front — gaps cost you eras
- ▶ Cluster specialty models by alpha theme, ensemble at the end
- ▶ Fresh-live refresh stays in the wrapper, not the model

Numerai Crypto & CrowdCent — the full crypto data-science loop

Daily Mon–Fri tournaments where every feature is one you sourced



Data sources & raw features

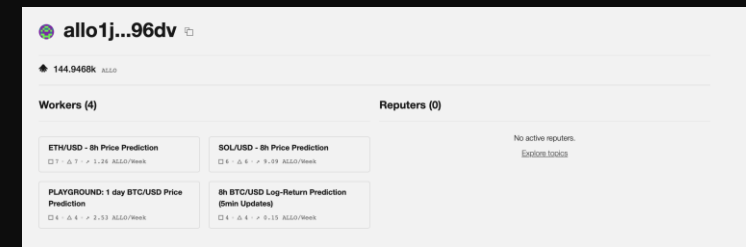
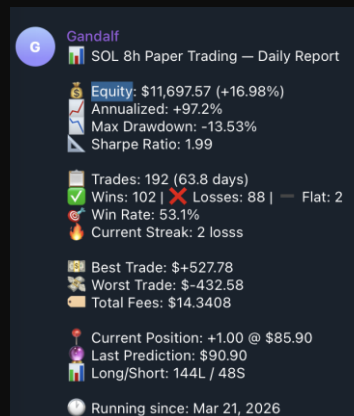
- Free crypto data APIs (CCXT, Binance, Coinalyze, etc.)
- Target assets + cross-asset universe
- OHLCV + quote volume + # trades
- Taker buy volume + taker buy quote volume
- Fear & Greed Index · DXY proxy · synthetic funding rate
- The more raw features the better

Feature engineering → model → backtest

- Feature engineering + statistical alpha analysis
- Regime-dependent discovery (bull/bear/chop)
- LightGBM / XGBoost / HGBR with bespoke tuning
- Metrics: CORR, MMC, hit-rate, Sharpe, drawdown
- Backtest yield analysis: multiple folds × timelines
- Policy network → PnL viz → paper-trade locally first

Allora Network

Decentralized intelligence — worker nodes submitting forecasts



What it is

- Workers submit inferences to on-chain topics
- Topics scored vs other workers in real time
- Wallet rewards for accurate, contributing models
- Multiple topics across BTC, ETH, SOL horizons

What we built — together

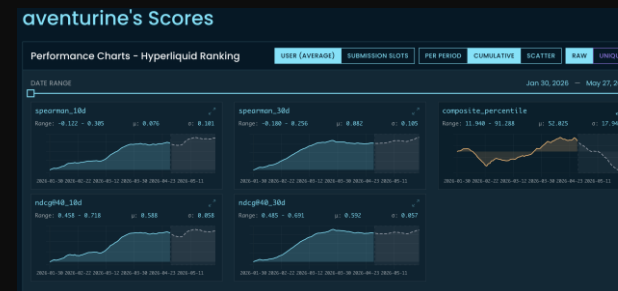
- Agent + I co-wrote a full research paper on the network
- Operated workers across 4+ topics — production scale
- Stood up a paper trading account running locally
- Built watchdog: process alive AND log mtime fresh
- Hardened against silent post-websocket deadlocks

Tips & tricks

- Research first — papers force you to actually understand the game
- Old watchdogs reported "OK" for 30 days while a topic was dead
- Paper trade locally before any vault deployment
- Always check log mtime + last submission timestamp
- pgrep alone is never sufficient — verify work is happening
- SIGTERM → 10s → SIGKILL → restart is the safe recovery

CrowdCent

Crowd-sourced model framework — multi-source crypto ranking



What it is

- Multiple model classes per horizon
- Daily submission slots, 5 per day
- Monthly retrain cadence (first Sunday)
- Staged publish protection built in

What we built — together

- 6 live models across two horizons
- Implemented the Advanced Multi-Source Ensemble pattern
- docs.crowdcent.com/tutorials/advanced-multi-data-slots/
- Monthly retrain cron with staged-publish protection
- flock prevents overlapping cron runs

Tips & tricks

- ▶ Ensemble across MULTIPLE data sources, not just multiple models
- ▶ Failed retrain must leave last-good model intact
- ▶ Stage retrain to candidate path → validate → swap, never in place
- ▶ flock + state files = no double-submits, no missed days

CrunchDAO — five competitions, five problem domains

Open data-science platform · each comp has its own project folder + memory notes

What CrunchDAO is

Open research competitions across biology, finance, time-series, and LLM tasks.
Each competition is a different problem; the agent's job is to learn the domain fast, then build a competitive model.

Broad Obesity 2

scRNA-seq perturbation prediction

Model: intensive-barnacle

ADIA Lab Structural Break

Detect structural breaks in
time-series under uncertainty

Model: solar-chicken

Synth

Probabilistic price forecasting
for crypto + stocks (CRPS scoring)

Models: molecular-cuckoo,
inquisitive-crow

DataCrunch 2

Equity return prediction
across thousands of moons

Model: dark-puma

Numinous

Open-ended LLM-augmented
research competition

Models: valuable-finch,
wild-kangaroo

Tips & tricks

- ▶ One project folder per comp — local memory notes per domain
- ▶ Sandbox never has your local env — guard OPENAI_API_KEY on LOCAL_DEV
- ▶ Agent does a deep-research pass on docs + leaderboard first
- ▶ crunch push ≠ deploy — promote manually in the hub UI

Every project is a self-driving research loop



- ▶ One scheduler · one state file · one report channel.
- ▶ Agent runs the loop; human runs the agent.
- ▶ Each new project is mostly copy-paste of the pattern.
- ▶ Failure modes are shared — fixing one upgrades all of them.

How to scope a pipeline with an agent

1. Frame

What is the competition rewarding? Write that in plain English first.

2. Scope

Smallest end-to-end pipeline that submits

3. Schedule

Decide cadence early. Cron makes it real.

4. State

state.json + idempotent reruns from day one.

5. Report

If you can't see it on Telegram, it didn't happen.

6. Iterate

Now improve the model. Not before.

Agent-native data science is the new default

- ▶ Numerai Skills + MCP — agents have native tournament access.
- ▶ Agent-to-agent: research agent $\leftrightarrow$ build agent $\leftrightarrow$ ops agent.
- ▶ Autonomous research loops running across multiple competitions.
- ▶ Open agents, open skills, open tournaments — DeAI all the way down.
- ▶ Your edge becomes: how you instrument and supervise the agent.

The bottleneck moves from "can you build a model" to "can you run a lab."

Thanks.

Questions?



@aventurine_eth



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Find me

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